
*
* STAAD.Pro V8i SELECTseries2 *
* Version 20.07.07.32 *
* Proprietary Program of *
* Bentley Systems, Inc. *
* Date= JAN 17, 2012 *
* Time= 13:35:55 *
*
* USER ID: RNS _ASSOCIATES *

1. STAAD SPACE
INPUT FILE: 20120117 JRC PIG STATION FLOOR JOIST.STD
2. START JOB INFORMATION
3. ENGINEER DATE 17-JAN-12
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 1 0 0 0; 2 3.45 0 0
9. MEMBER INCIDENCES
10. 1 1 2
11. DEFINE MATERIAL START
12. ISOTROPIC STEEL
13. E 2.05E+008
14. POISSON 0.3
15. DENSITY 76.8195
16. ALPHA 1.2E-005
17. DAMP 0.03
18. TYPE STEEL
19. STRENGTH FY 275000 FU 407800 RY 1.5 RT 1.2
20. END DEFINE MATERIAL
21. MEMBER PROPERTY JAPANESE
22. 1 TABLE ST C100X50X5
23. CONSTANTS
24. MATERIAL STEEL ALL
25. SUPPORTS
26. 1 PINNED
27. 2 FIXED BUT MY MZ KMX 0.00099
28. LOAD 1 LOADTYPE NONE TITLE DEAD LOAD: DL
29. SELFWEIGHT Y -1
30. MEMBER LOAD
31. 1 UNI GY -0.25
32. LOAD 2 LOADTYPE NONE TITLE LIVE LOAD: LL
33. MEMBER LOAD
34. 1 UNI GY -2.5
35. LOAD COMB 3 1.0DL + 1.0LL
36. 1 1.0 2 1.0
37. LOAD COMB 4 1.4DL + 1.6LL
38. 1 1.4 2 1.6
39. PERFORM ANALYSIS

PROBLEM STATISTICS

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 2/ 1/ 2

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 2, TOTAL DEGREES OF FREEDOM = 6
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 366909.1 MB

40. PRINT SECTION DISPL ALL

MEMBER SECTION DISPLACEMENTS

UNITS ARE - CM

MEMB	LOAD	GLOBAL X,Y,Z DISPL FROM START TO END JOINTS AT 1/12TH PTS					
1	1	0.0000	0.0000	0.0000	0.0000	-0.0427	0.0000
		0.0000	-0.0822	0.0000	0.0000	-0.1158	0.0000
		0.0000	-0.1413	0.0000	0.0000	-0.1572	0.0000
		0.0000	-0.1626	0.0000	0.0000	-0.1572	0.0000
		0.0000	-0.1413	0.0000	0.0000	-0.1158	0.0000
		0.0000	-0.0822	0.0000	0.0000	-0.0427	0.0000
	2	0.0000	0.0000	0.0000	0.0000	-0.3128	0.0000
		0.0000	-0.6020	0.0000	0.0000	-0.8476	0.0000
		0.0000	-1.0341	0.0000	0.0000	-1.1504	0.0000
		0.0000	-1.1899	0.0000	0.0000	-1.1504	0.0000
		0.0000	-1.0341	0.0000	0.0000	-0.8476	0.0000
		0.0000	-0.6020	0.0000	0.0000	-0.3128	0.0000
	3	0.0000	0.0000	0.0000	0.0000	-0.3555	0.0000
		0.0000	-0.6842	0.0000	0.0000	-0.9634	0.0000
		0.0000	-1.1754	0.0000	0.0000	-1.3076	0.0000
		0.0000	-1.3525	0.0000	0.0000	-1.3076	0.0000
		0.0000	-1.1754	0.0000	0.0000	-0.9634	0.0000
		0.0000	-0.6842	0.0000	0.0000	-0.3555	0.0000
	4	0.0000	0.0000	0.0000	0.0000	-0.5603	0.0000
		0.0000	-1.0783	0.0000	0.0000	-1.5183	0.0000
		0.0000	-1.8523	0.0000	0.0000	-2.0607	0.0000
		0.0000	-2.1315	0.0000	0.0000	-2.0607	0.0000
		0.0000	-1.8523	0.0000	0.0000	-1.5183	0.0000
		0.0000	-1.0783	0.0000	0.0000	-0.5603	0.0000

MAX LOCAL DISP = 2.13151 AT 172.50 LOAD 4 L/DISP= 161

***** END OF SECT DISPL RESULTS *****

41. PRINT MEMBER FORCES

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN METE (Local)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
1	1	1	0.00	0.59	0.00	0.00	0.00	0.00
		2	0.00	0.59	0.00	0.00	0.00	0.00
	2	1	0.00	4.31	0.00	0.00	0.00	0.00
		2	0.00	4.31	0.00	0.00	0.00	0.00
	3	1	0.00	4.90	0.00	0.00	0.00	0.00
		2	0.00	4.90	0.00	0.00	0.00	0.00
	4	1	0.00	7.72	0.00	0.00	0.00	0.00

***** END OF LATEST ANALYSIS RESULT *****

42. PARAMETER 1
43. CODE BS5950
44. TRACK 2 ALL
45. BEAM 2 ALL
46. CHECK CODE ALL

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.12_5950-1_2000

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
* 1	ST C100X50X5	FAIL 0.00	BS-4.3.6 0.00	1.229 6.66	4 0.00

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 275 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 345.00
Gross Area = 11.92 Net Area = 11.92 Eff. Area = 11.92

	z-z axis	y-y axis
Moment of inertia	188.000	26.000
Plastic modulus	44.300	15.300
Elastic modulus	37.600	7.514
Effective modulus	44.300	15.300
Shear Area	6.750	5.000

DESIGN DATA (units - kN,m)

Section Class : PLASTIC

	z-z axis	y-y axis
Moment Capacity	12.2	3.1
Reduced Moment Capacity	12.2	3.1
Shear Capacity	111.4	82.5

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

LTB Moment Capacity (kNm) and LTB Length (m): 5.42, 3.450
LTB Coefficients _Associated Moments (kNm):
mLT = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
MLt = 6.66 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.2.3-(Y)	0.094	4	-	7.7	-	-	-
BS-4.3.6	1.229	4	-	7.7	-	6.7	-

Torsion and deflections have not been considered in the design.

***** END OF TABULATED RESULT OF DESIGN *****

47. FINISH

***** END OF THE STAAD.Pro RUN *****

**** DATE= JAN 17,2012 TIME= 13:35:55 ****

* For questions on STAAD.Pro, please contact *
* Bentley Systems or Partner offices *
* *
* Telephone Web / Email *
* USA +1 (714) 974-2500 *
* UK +44 (0) 808 101 9246 *
* SINGAPORE +65 6225-6158 *
* FRANCE +33 (0) 1 55238400 *
* GERMANY +49 0931 40468 *
* INDIA +91 (033) 4006-2021 *
* JAPAN +81 (03)5952-6500 http://www.ctc-g.co.jp *
* CHINA +86 21 6288 4040 *
* THAILAND +66 (0)2645-1018/19 partha.p@reisoftwareth.com*
* *
* Worldwide http://selectservices.bentley.com/en-US/ *
* *
